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(54) **HYPERSPECTRAL SONAR**

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Aug. 30, 2018, now Pat. No. 11,054,521, which is a
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Primary Examiner — James R Hulka

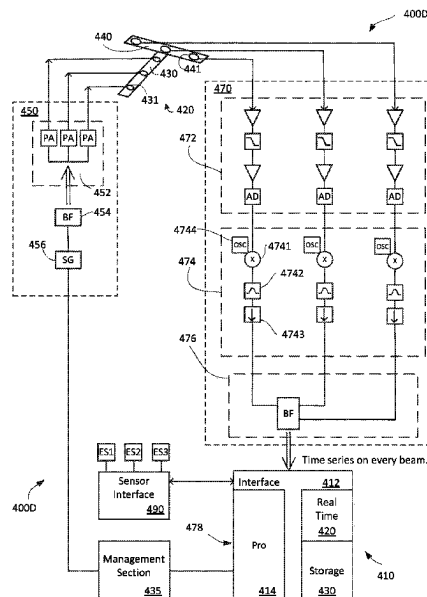
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Ocean Law

(57)

ABSTRACT

A sonar survey system and method excites reflectors using
a broadband message that excites all frequencies within the
band providing for selection and evaluation of frequencies
of interest after the survey is completed.

17 Claims, 14 Drawing Sheets



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FIG. 1

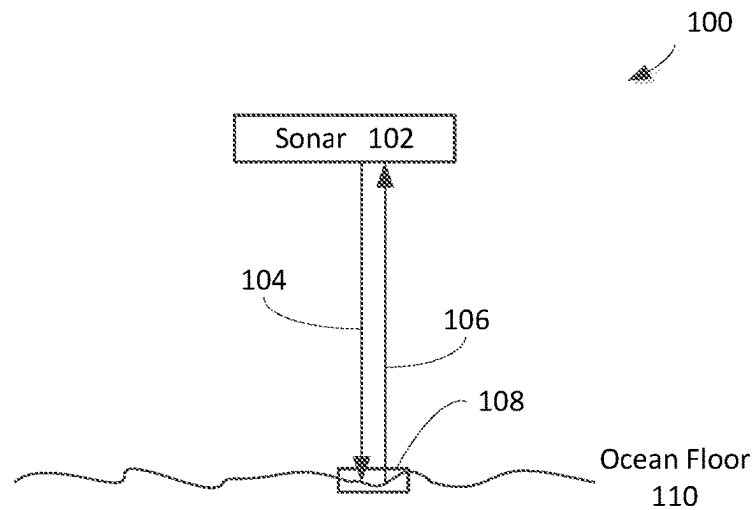


FIG. 2A

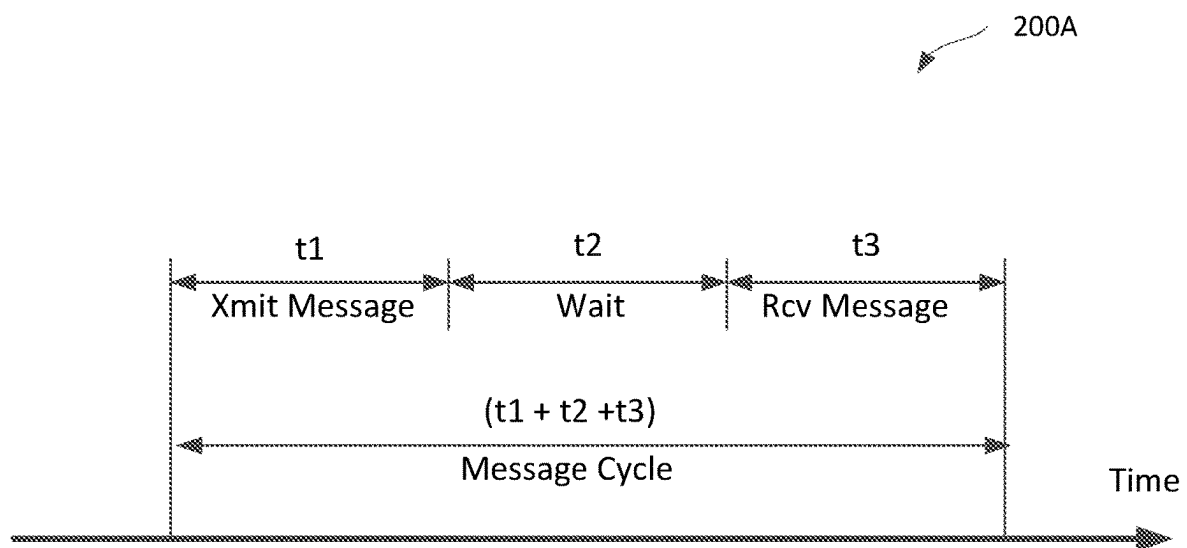


FIG. 2B

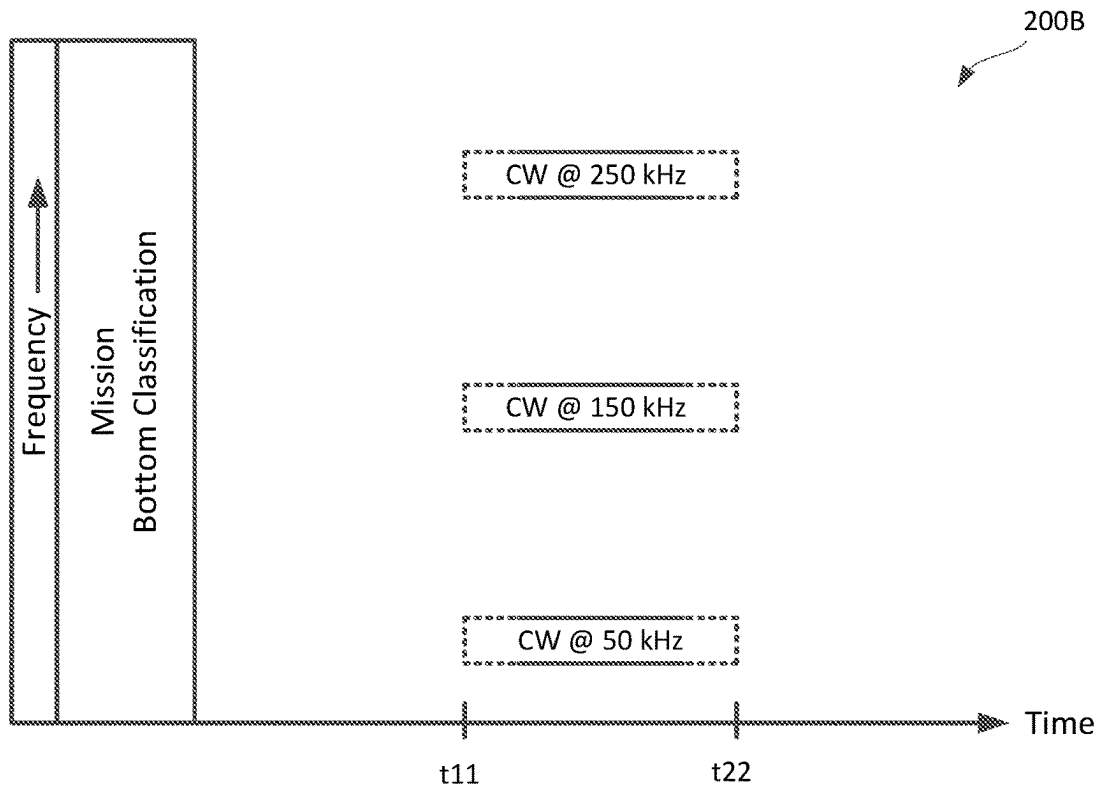


FIG. 2C

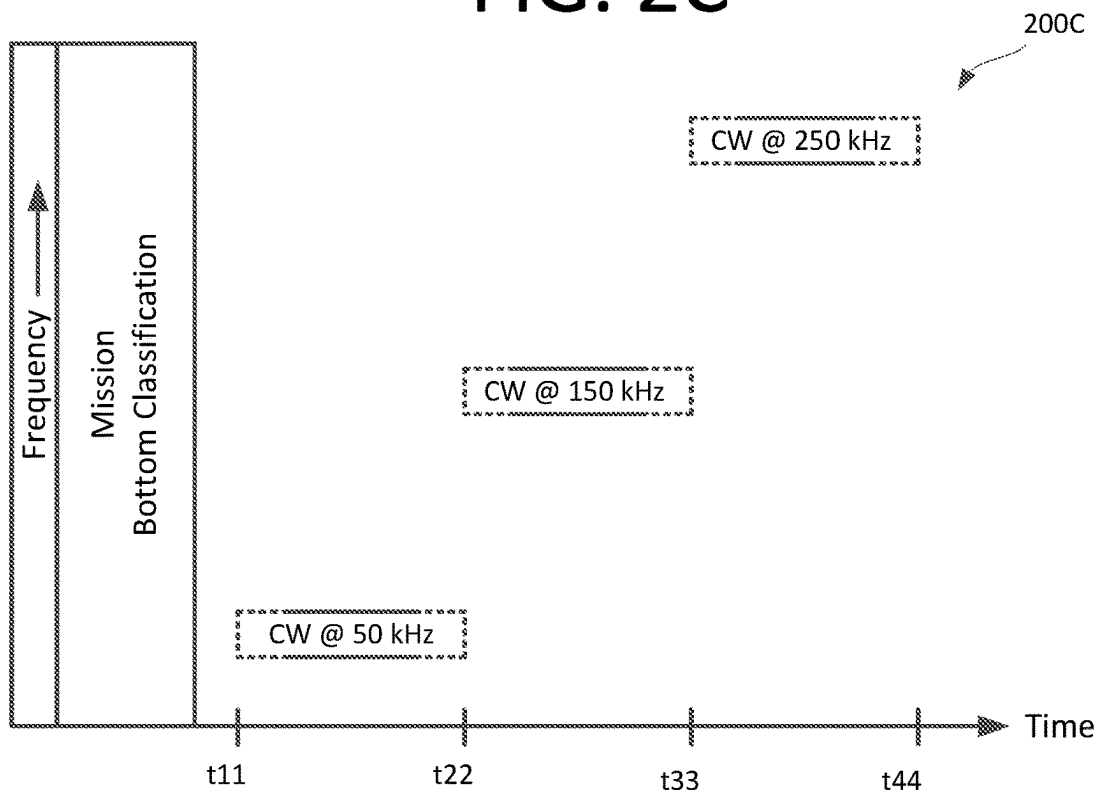


FIG. 2D

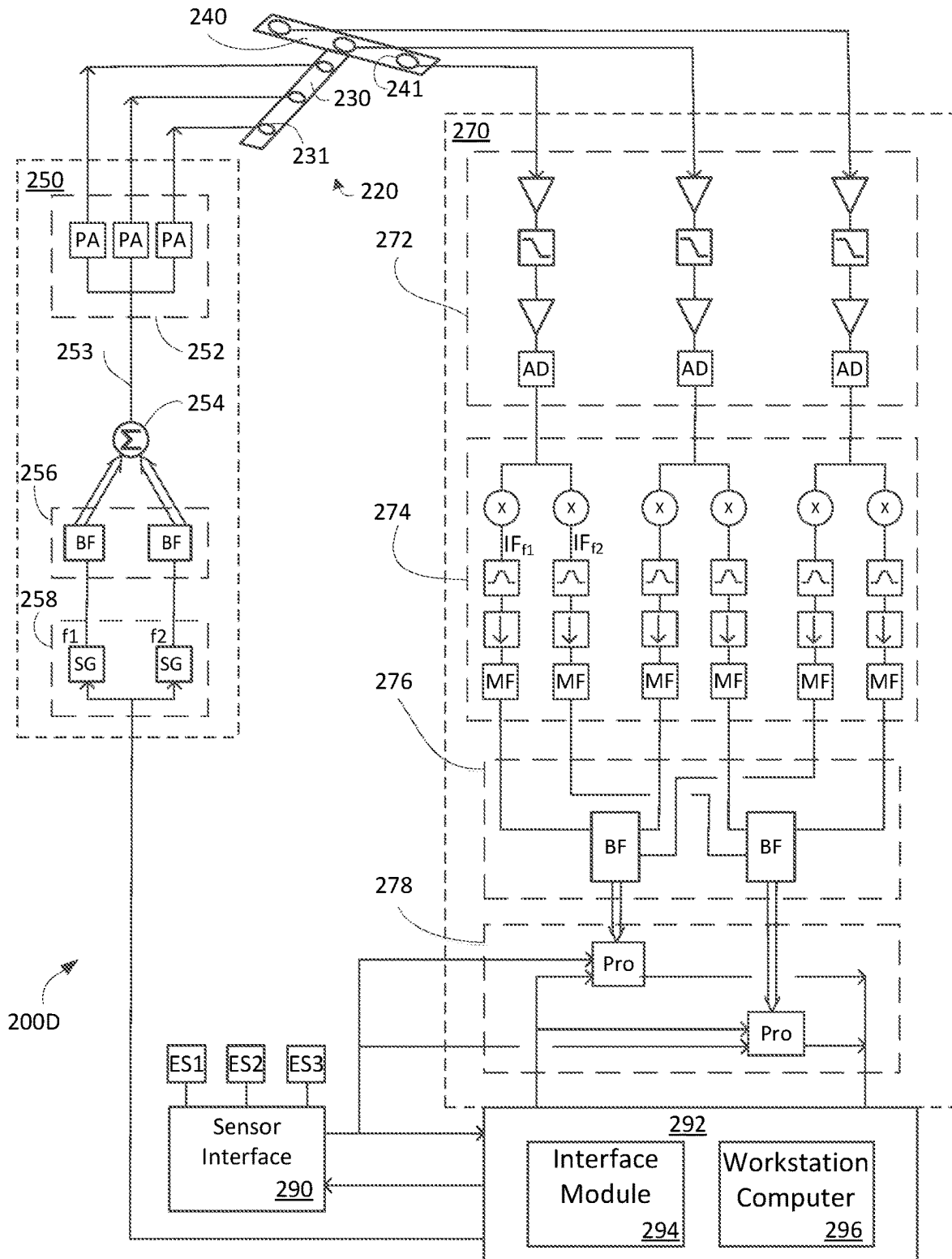


FIG. 3A

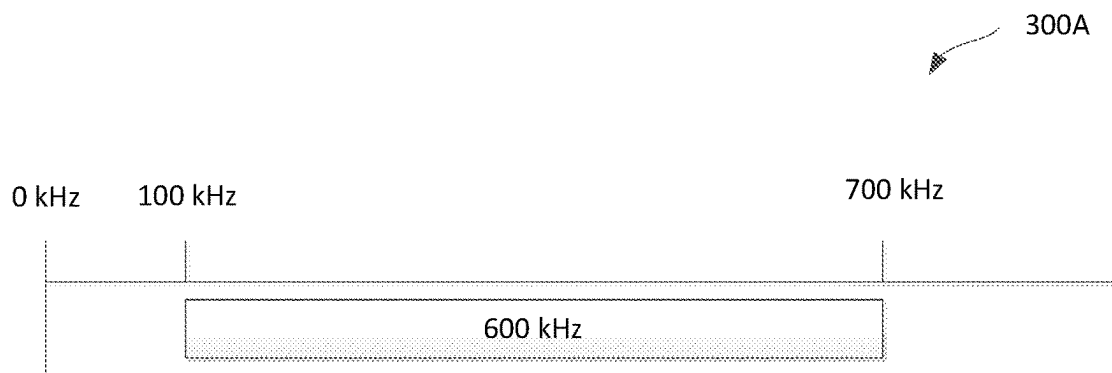


FIG. 3B

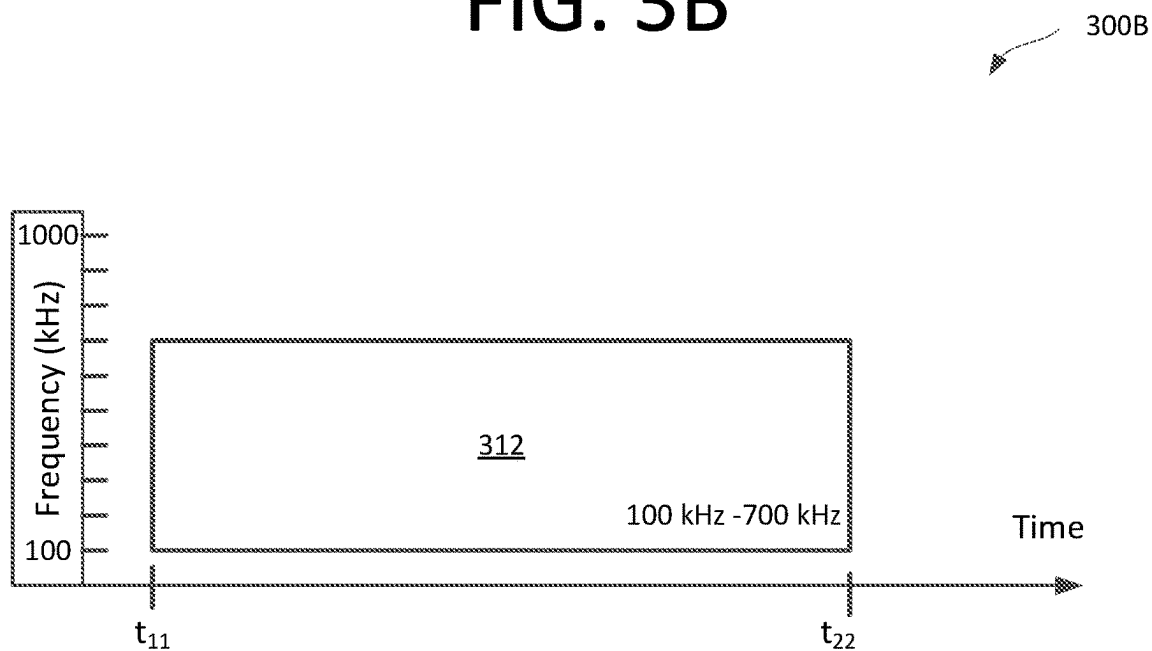


FIG. 3C

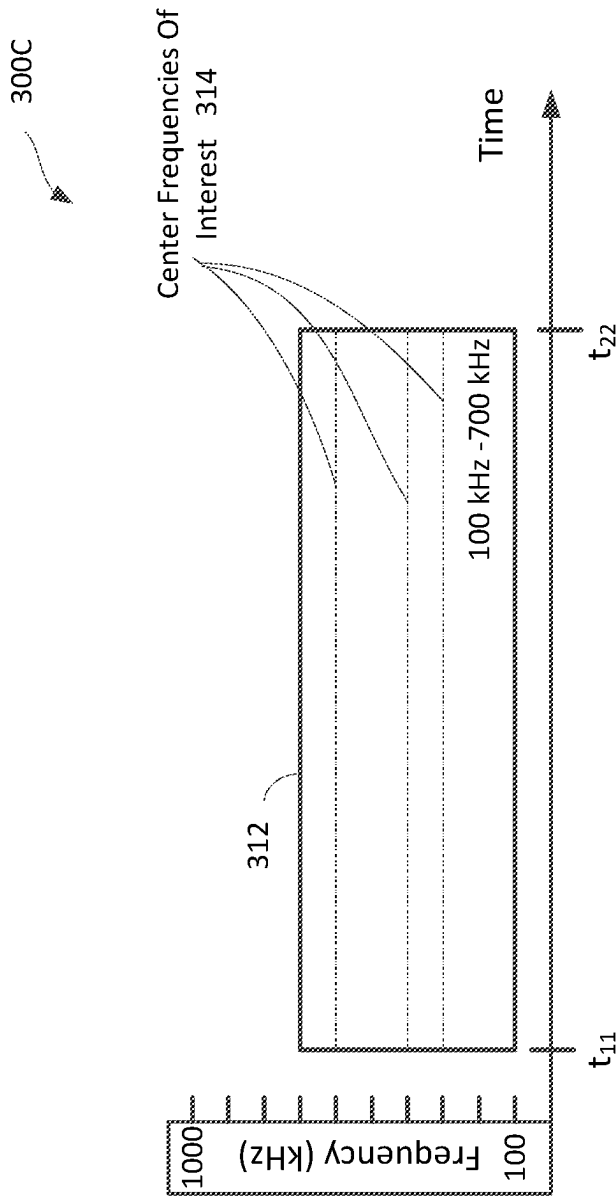


FIG. 4A

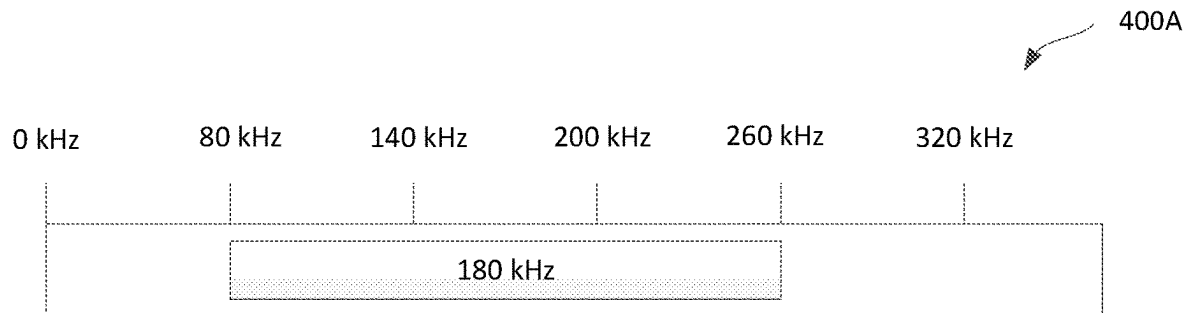


FIG. 4B

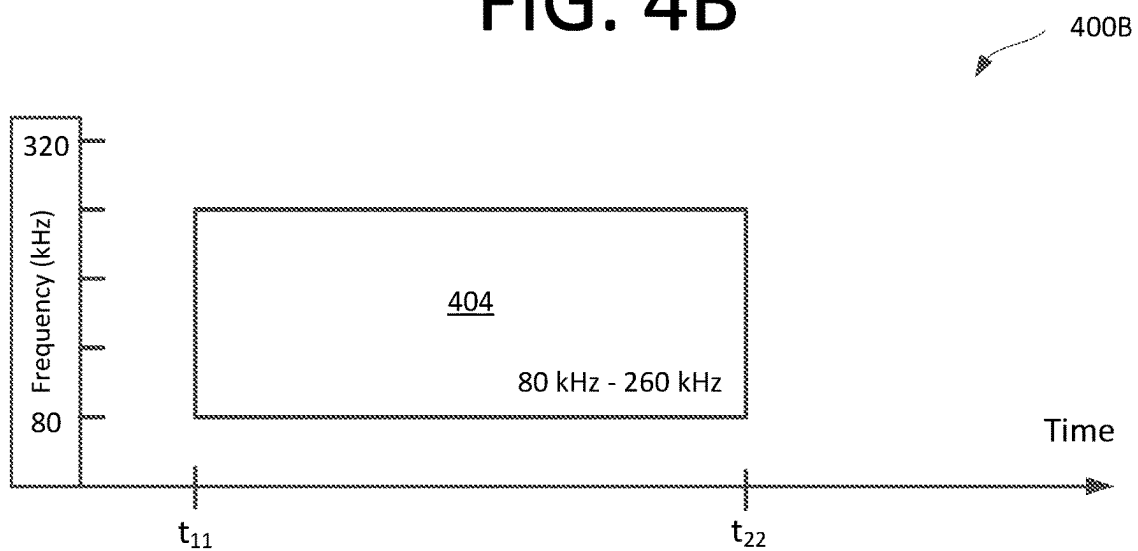


FIG. 4C

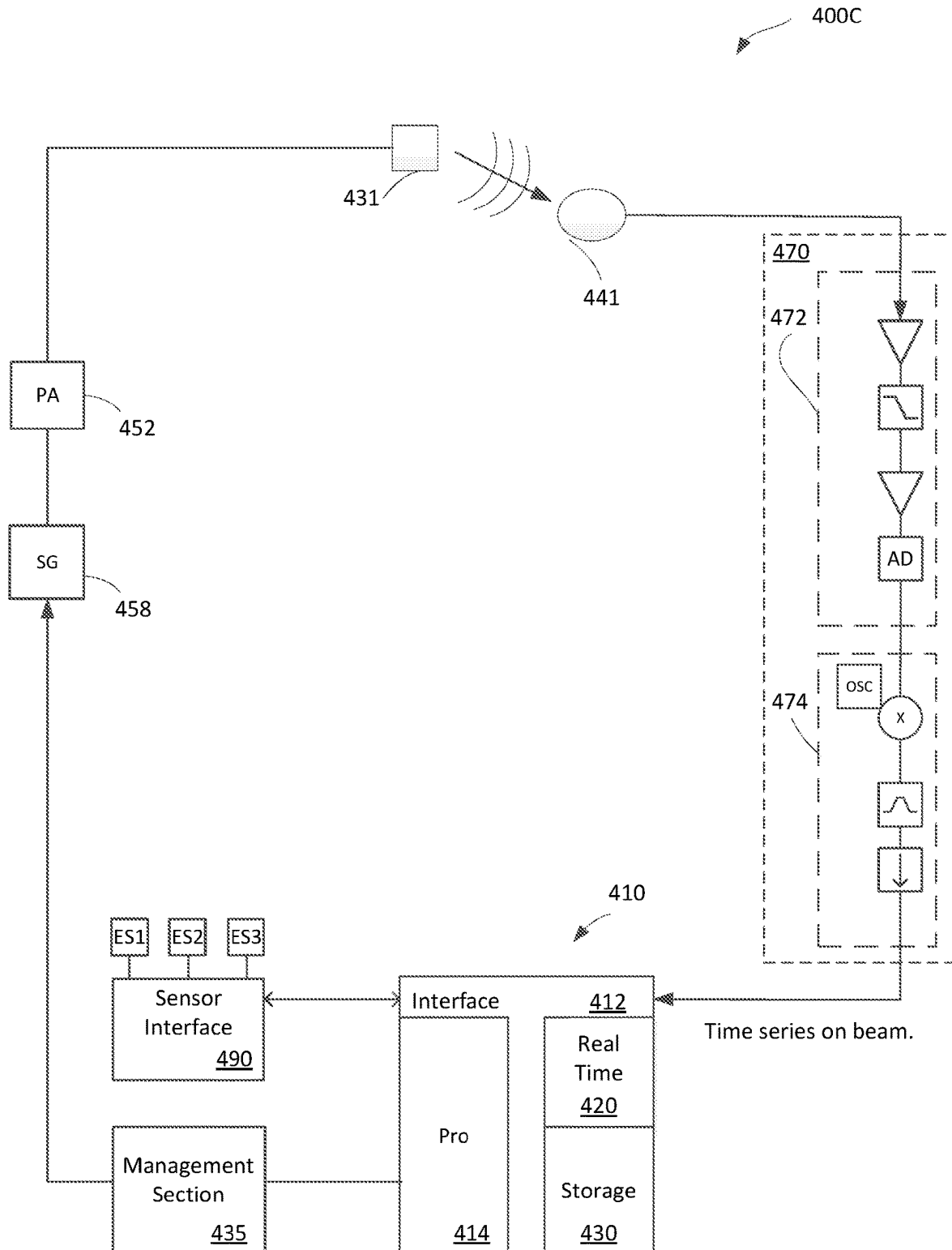


FIG. 4D

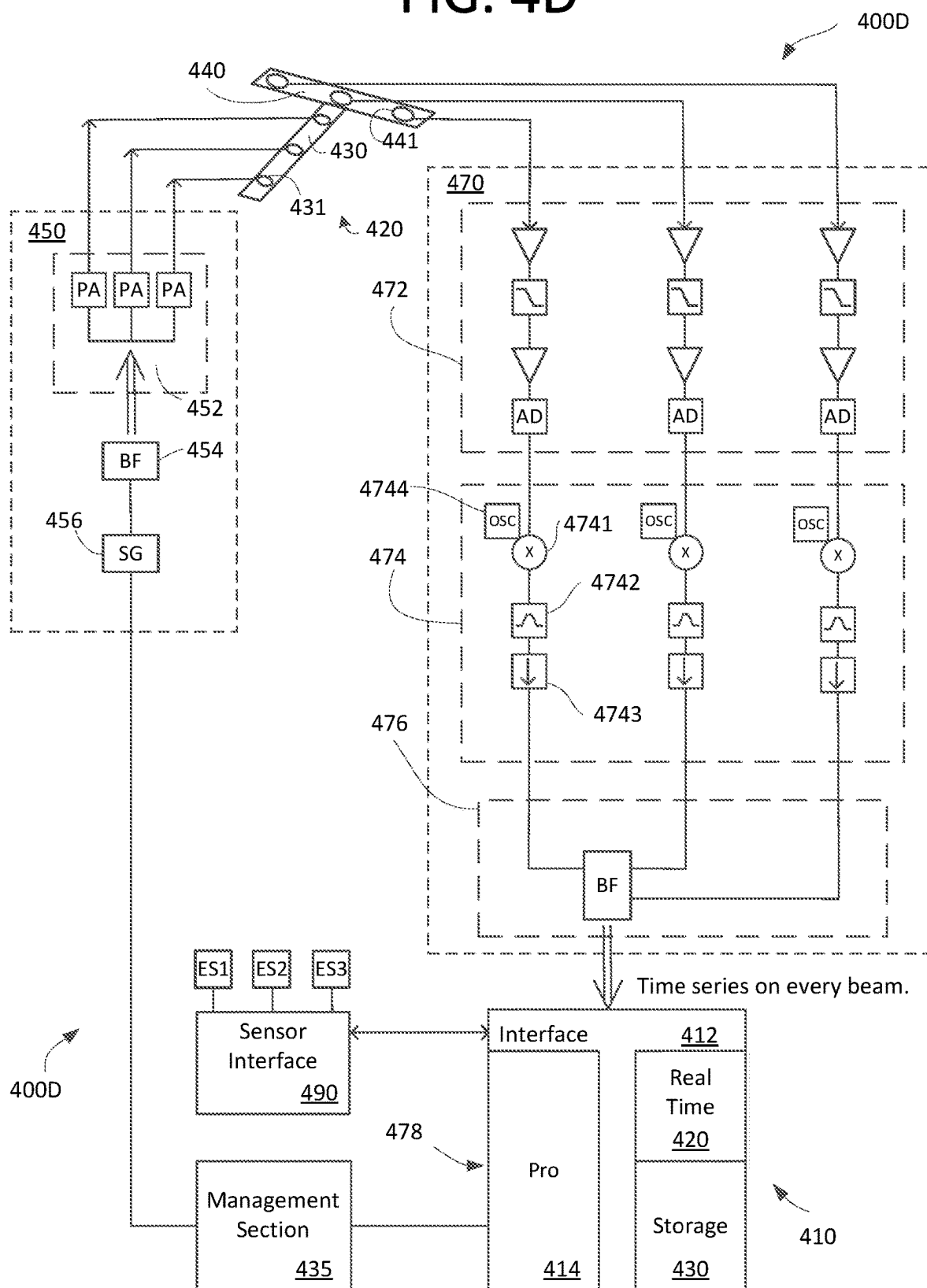


FIG. 4E

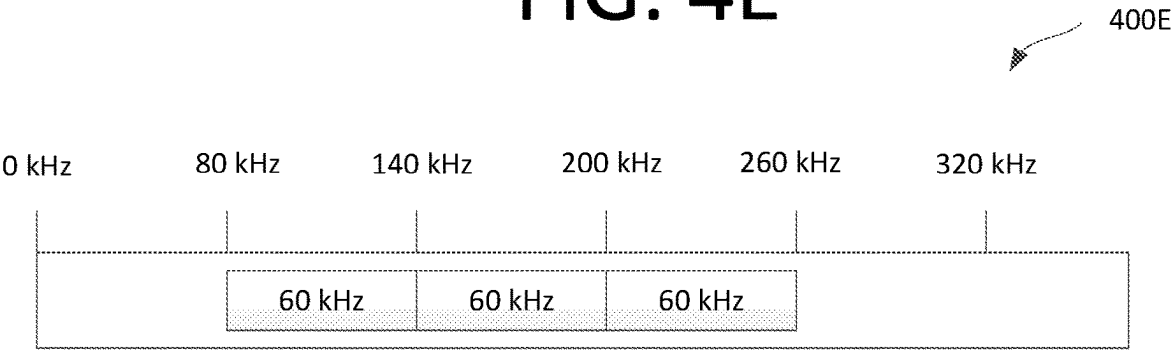


FIG. 4F

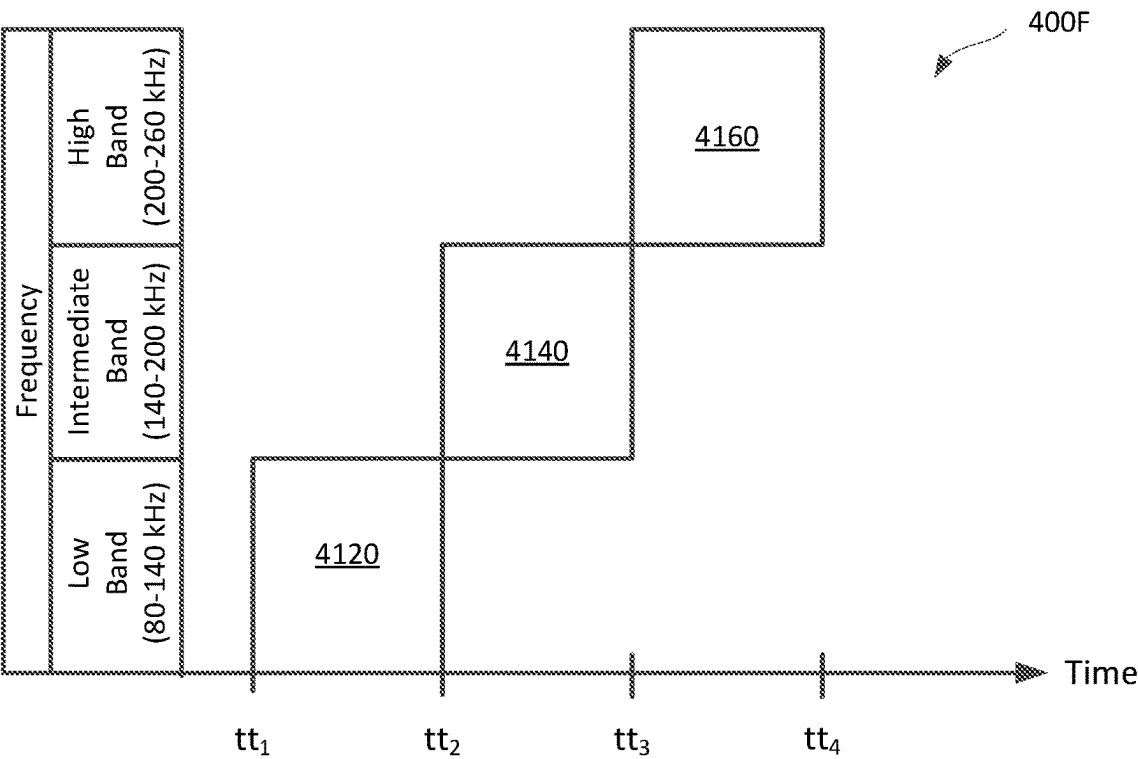


FIG. 4G

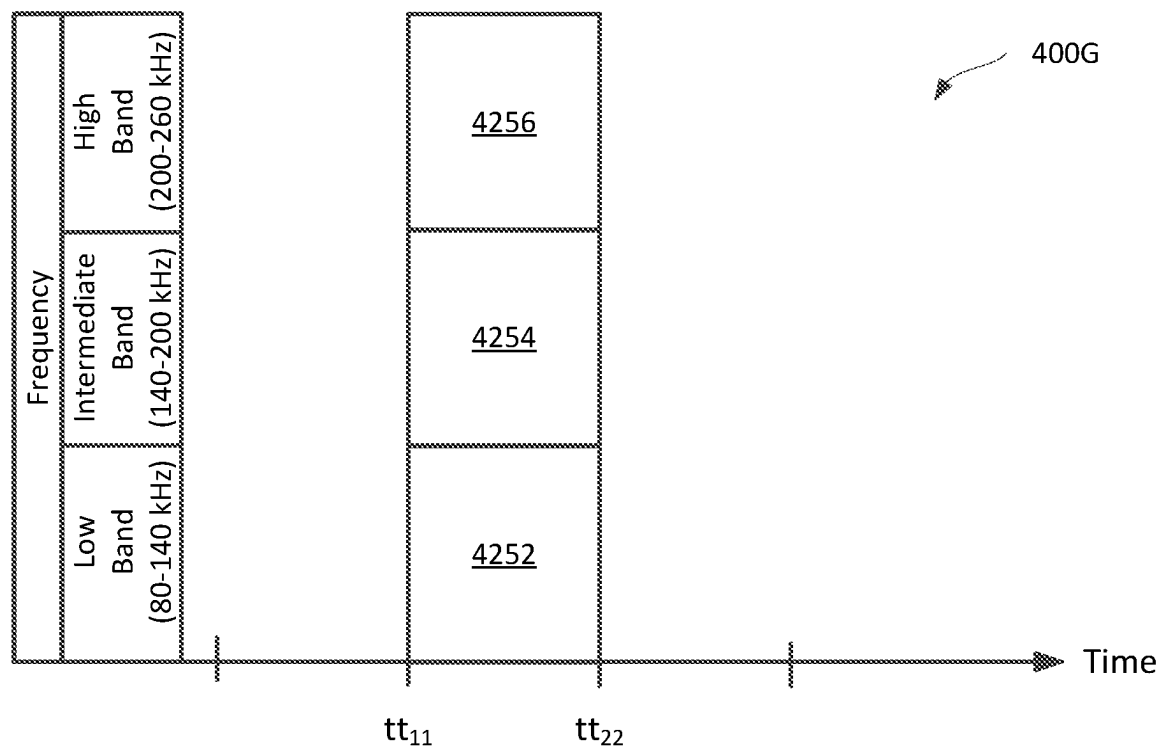


FIG. 4H

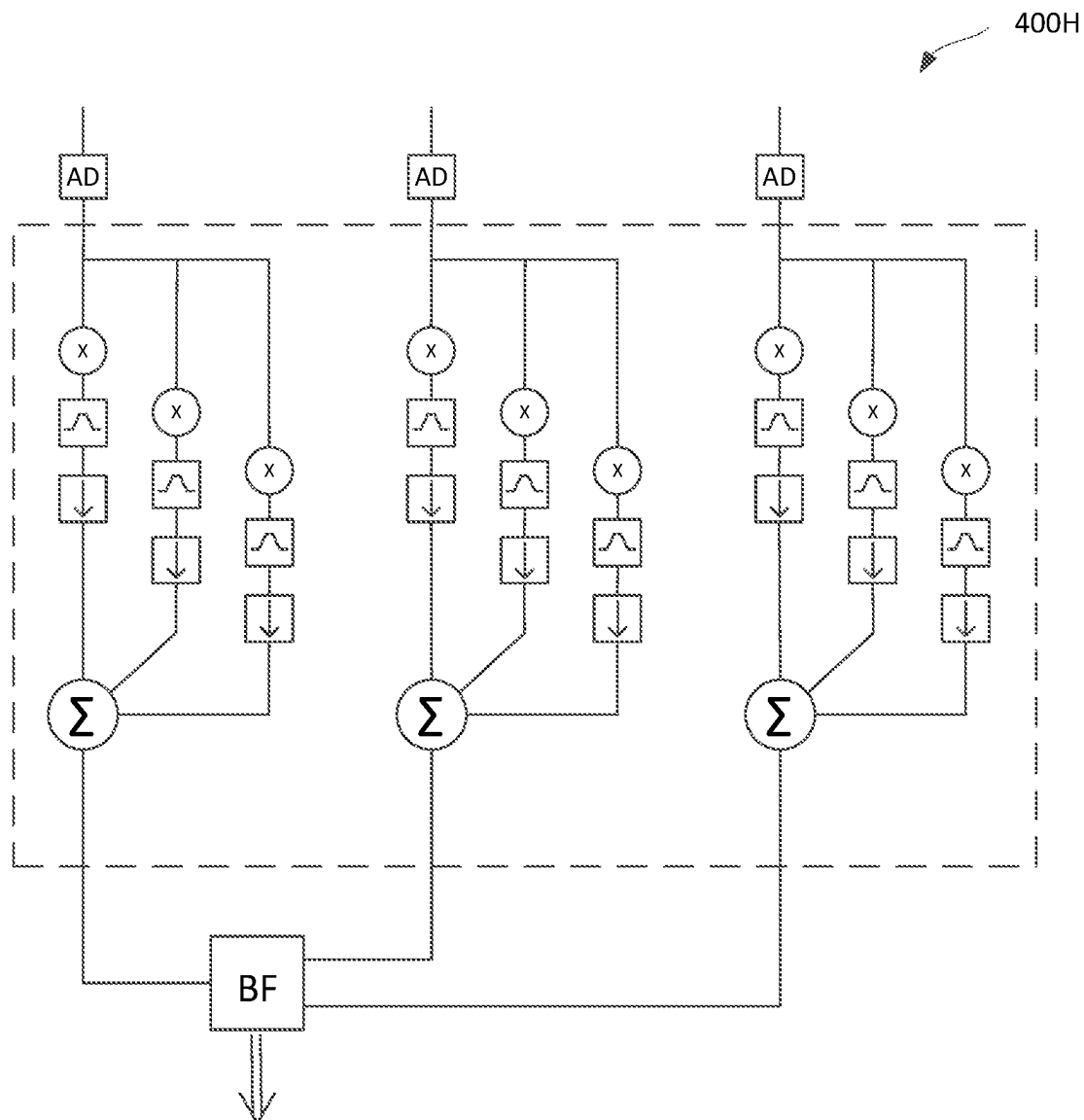


FIG. 5

500

HYPER SPECTRAL MISSION	SIGNAL TYPE			
	FM	Noise-like	Click	Click Train
1) Bottom classification and/or segmentation	X	X	X	X
2) Sub-bottom classification and/or segmentation	X	X	X	X
3) Improved bottom detection	X	X	X	X
4) Water column classification and/or segmentation	X	X	X	X
5) Object classification	X	X	X	X

FIG. 6A

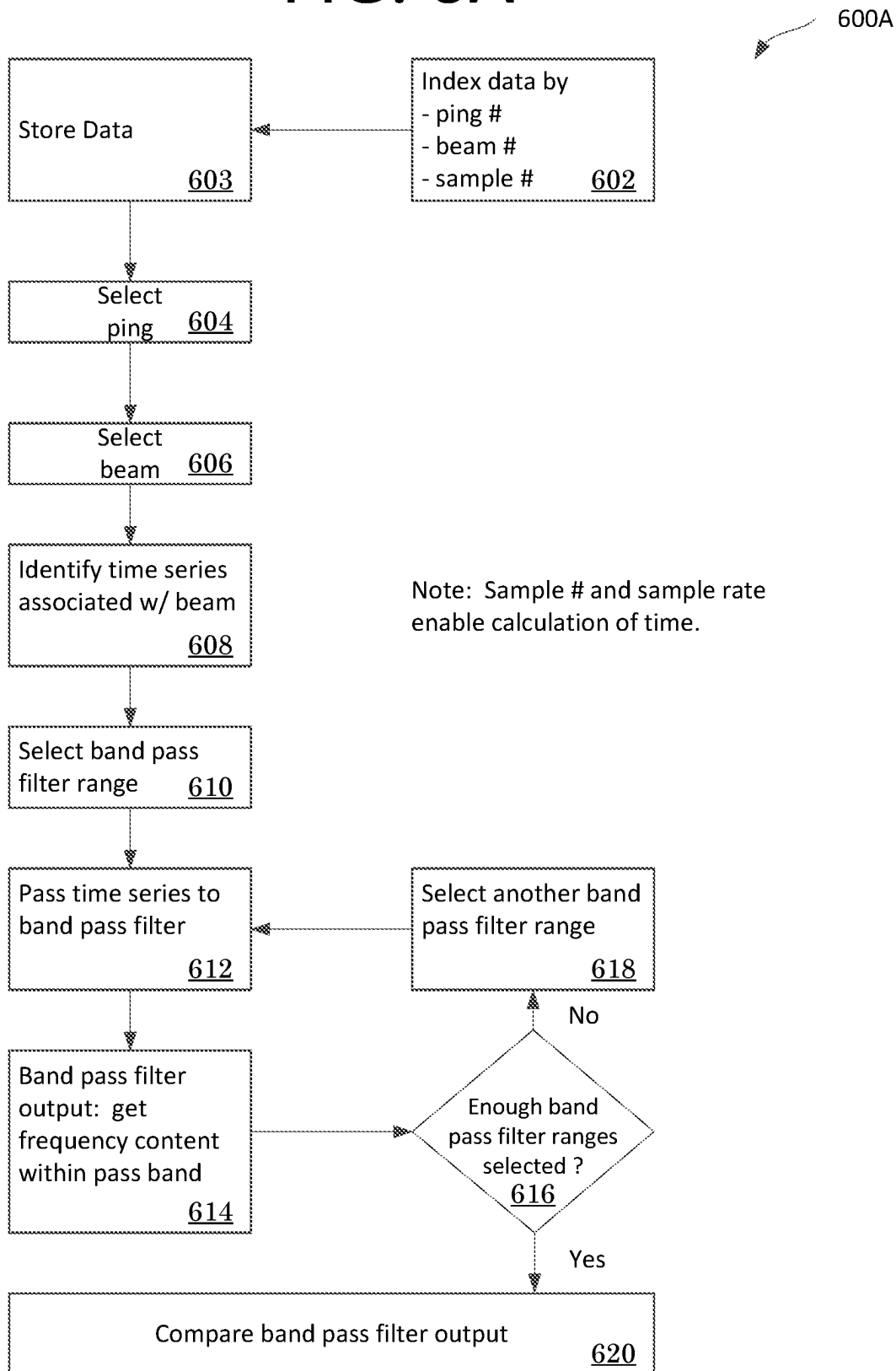
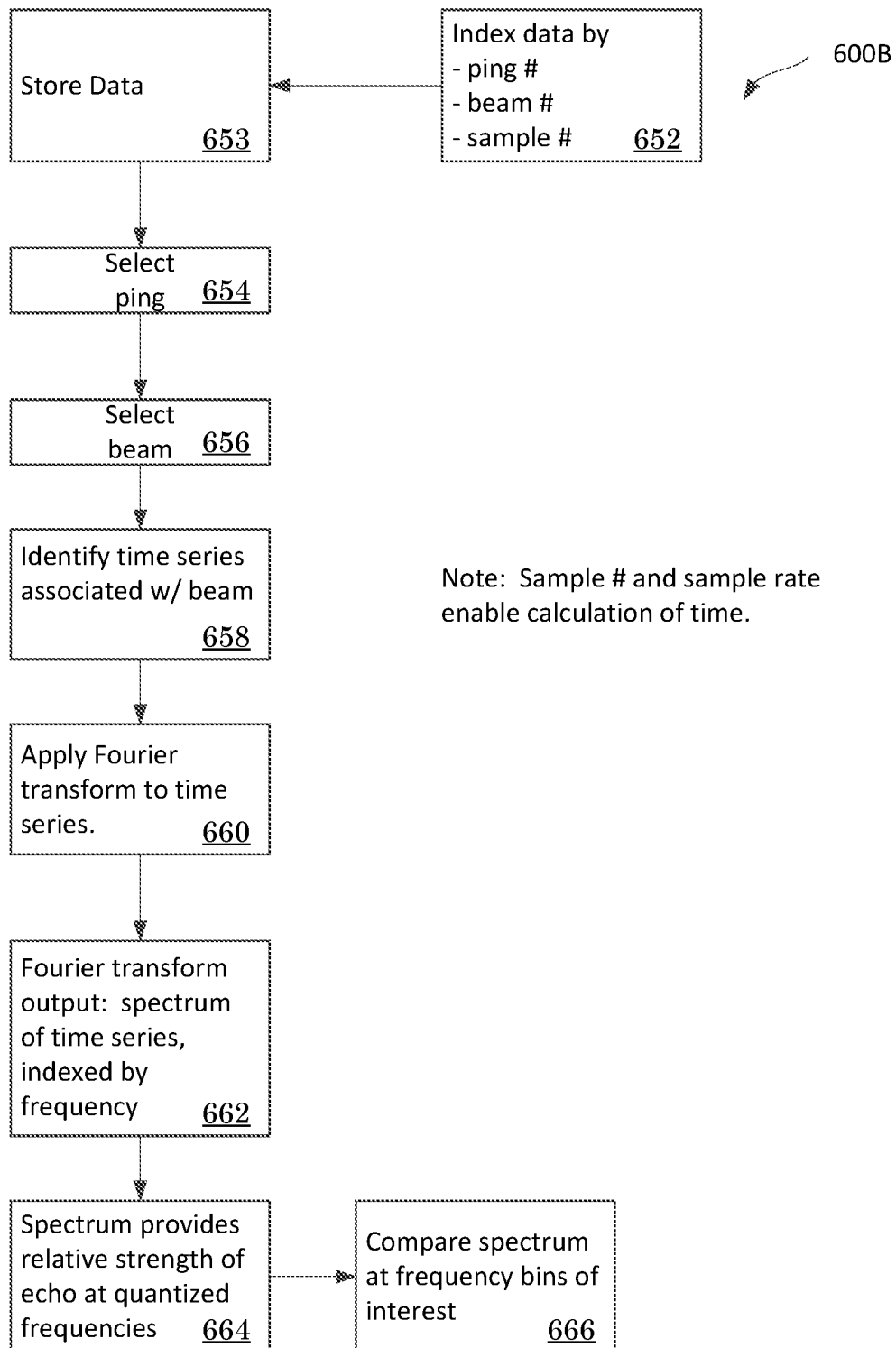


FIG. 6B



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HYPERSPECTRAL SONAR**PRIORITY APPLICATIONS AND
INCORPORATION BY REFERENCE**

This application is a continuation of Ser. No. 17/334,693 filed May 29, 2021 entitled Hyperspectral Sonar which is a continuation of Ser. No. 16/118,327 filed Aug. 30, 2018 entitled Hyperspectral Sonar which is a continuation of Ser. No. 15/700,482 filed Sep. 11, 2017 entitled Hyperspectral Sonar, now U.S. Pat. No. 10,067,228. This application incorporates by reference, in their entireties and for all purposes, the disclosures of U.S. Pat. No. 3,144,631 concerning Mills Cross sonar, U.S. Pat. No. 8,305,841 concerning sonar used for mapping seafloor topography and U.S. Pat. No. 3,488,445 concerning matched filtering of orthogonal signals.

BACKGROUND OF THE INVENTION**Field of the Invention**

The present invention relates to underwater acoustical systems, methods for using underwater acoustical systems, and methods for processing and using the data they produce. In particular, the invention relates to hyperspectral survey systems.

Discussion of the Related Art

Sonar survey systems that operate at one or at a few frequencies are known in applications including bathymetric surveys and bottom classification. However, although broadband survey systems can also be used for bathymetric surveys, their use for sea floor and water column classification are generally unknown.

SUMMARY OF THE INVENTION

The present invention provides a hyperspectral sonar system. With goals similar to bottom characterization and water column characterization missions, hyperspectral sonar uses substantially different transmit waveforms, beamforming, and post-processing methods to acquire data across a broad spectrum of frequencies.

Hyperspectral operation can be contrasted with multi-spectral techniques. In the one, multi-spectral, acoustic data are acquired at particular frequencies such that results are available only at those frequencies in use during the original data acquisition. Frequencies of interest to the user must be known and chosen ahead of time prior to data collection. In the other, hyperspectral, data are acquired across a broad frequency range. Contemporaneous or post processed hyperspectral data therefore provide results at any frequencies of interest up to the Nyquist frequency of the system with a frequency resolution of $1/\text{acquisition time}$. Compared to the multi-spectral case, hyperspectral techniques offer higher frequency resolution when collecting frequency-dependent results and provide the user flexibility in choosing frequencies of interest in post-processing; no prior knowledge is required.

Like the multi-spectral system, the minimum and maximum operating frequencies of a hyperspectral system are determined by the operating band of the sonar. However, unlike multi-spectral systems that utilize multiple sonar systems for transmitting multiple waveforms, hyperspectral sonar has no such requirement; a single hyperspectral sys-

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tem will suffice. Furthermore, no library of transmit waveforms with different center frequencies is required since one broadband waveform may substitute for all.

Applicant notes that some known sonar systems may utilize broadband signals. However, these systems do not operate like applicant's hyperspectral sonar system. For example, these systems may have one or more of the following characteristics which distinguish them from embodiments of applicant's invention:

- operate at low frequencies less than 20 kHz;
- have a relatively small bandwidth spanning less than a decade of frequencies;
- be single beam systems with one broad beam of around 10 degrees beamwidth in one dimension;
- be oriented such that beam(s) are horizontal;
- use pulse compression or matched filtering techniques;
- and classify targets based on resonance.

Take, for example, broadband sonar systems used in fishing applications. Such systems possess many of the above qualities and are incompatible with applicant's hyperspectral sonar system. A fisheries sonar typically uses a wide beam to integrate the combined backscatter over a large volume of water for estimating total biomass in the volume, and interpretation of the local minima and maxima of the returned spectrum allows for species estimation based on known resonances of fish. To contrast, a hyperspectral system focuses a very narrow beam (for example, $1^\circ \times 1^\circ$ beamwidth) on discrete targets, making it impractical for integrations over large volumes. Hyperspectral systems also exclude matched filters and examine echo content at only specific frequencies of interest.

In an embodiment, applicant's survey system provides: a vertical survey system including a broadband multibeam echo sounder system that avoids the use of matched filters for installation on a water going vehicle; an acoustic transceiver for use with one or more transducers in a single projector array and plural transducers in a single hydrophone array; the projector array arranged with respect to the hydrophone array to form a Mills Cross; the system capable of forming beams with $1^\circ \times 1^\circ$ or better resolution; a transceiver for synthesizing a transmitter message including a frequency modulated (FM), noise-like, click, or click train message with a frequency between 20 and 1000 kHz and with a bandwidth of 100 kHz or more; and, the message for exciting the projector array such that a 180 degree or smaller swath of a waterbody bottom is ensonified by the message, and a message echo from ensonified scattering centers is returned to the hydrophone array; wherein data derived from the hydrophone returns are stored and subsequently made available for analysis of any frequency within the band using narrow band pass filters or spectral analysis and comparison of the results from any two or more frequencies.

In an embodiment, applicant's survey system provides: A method of comparing band pass filter outputs to assess the strength of echo returns at particular frequencies, the method comprising the steps of acquiring multibeam sonar data and indexing the data by ping number, beam number, and sample number; storing the sonar data; selecting a ping and selecting a beam of the ping; processing one beam at a time by identifying a time series associated with the beam, using no matched filters, selecting a band pass filter centered at a discrete frequency f_i where the selected band pass filter frequency need not match the frequency of the transmitted waveform, processing the time series through a band pass filter centered at discrete frequency f_i , and repeating the

processing step for n frequencies where n is two or more; and, comparing the band pass filter outputs at two or more band pass filter frequencies.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention is described with reference to the accompanying figures. These figures, incorporated herein and forming part of the specification, illustrate embodiments of the invention and, together with the description, further serve to explain its principles enabling a person skilled in the relevant art to make and use the invention.

FIG. 1 shows a sonar system of the present invention **100**.

FIG. 2A shows a message timing diagram **200A**.

FIG. 2B-C show multispectral messages in parallel and serial format **200B-C**.

FIG. 2D shows a multispectral survey system **200D**.

FIGS. 3A-C show hyperspectral messages and frequencies of interest **300A-C**.

FIGS. 4A-B show hyperspectral messages in an 80 to 260 kHz band **400A-B**.

FIGS. 4C-D show hyperspectral survey systems **400C-D**.

FIGS. 4E-G show hyperspectral messages and sub-bands **400E-G**.

FIG. 4H shows a modification to the survey system of FIG. 4D enabling operation with sub-bands **400H**.

FIG. 5 shows hyperspectral mission types **500**.

FIG. 6A shows a hyperspectral survey process using band pass filters **600A**.

FIG. 6B shows a hyperspectral survey process using Fourier analysis **600B**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The disclosure provided in the following pages describes examples of some embodiments of the invention. The designs, figures, and description are non-limiting examples of the embodiments they disclose. For example, other embodiments of the disclosed device and/or method may or may not include the features described herein. Moreover, described features, advantages or benefits may apply to only certain embodiments of the invention and should not be used to limit the disclosed invention.

As used herein, the term "coupled" includes direct and indirect connections. Moreover, where first and second devices are coupled, intervening devices including active devices may be located therebetween.

Frequency Dependence of Sonar Returns

FIG. 1 shows a sonar system **102** that ensonifies a location on an ocean floor **108**. The sonar system **102** emits an acoustic signal or message **104** from projector transducers and the signal strikes scattering centers or reflectors **108** at the ocean floor **110**. Returns from the acoustic interaction **106** are received by hydrophone transducers **107** at the sonar **102**.

The acoustic interaction between the emitted signal and the reflector produces a return that depends on a number of variables including angle of incidence and frequency of the emitted signal. For example, when the survey system is directly over the reflector the backscatter strength may be a maximum with lesser angles of incidence having less backscatter strength.

Seabed conditions may be of interest. In the case of varying frequency, at low frequencies where acoustic wavelengths are larger than the scale of seabed roughness, the seabed surface may appear to be acoustically smooth. On the

other hand, at high frequencies such that acoustic wavelengths are smaller than the scale of seabed roughness, scattering can dominate the returning signal and the seabed may be considered acoustically rough. Water column conditions may be of interest. Here, low frequencies with large acoustic wavelengths may be required to match the scale of water column features. At this scale, water column features such as fish come to mind. Where water column features such as particulate, small bubbles, and plankton are of interest, higher frequencies with smaller acoustic wavelengths may be required to match the scale of the feature.

Where conditions of the ocean bottom or water column suggest that the response of reflectors to disparate frequencies will be different owing for example to differing hardness (acoustic reflection coefficient), roughness (as a backscatter coefficient) or scale of an obstruction, multispectral sonar systems may be used to distinguish among types of bottoms or water column obstructions encountered.

Multispectral Sonar Systems

FIG. 2A shows a message cycle **200A**. In particular, the sonar system emits a single ping acoustic message during **t1**. And, during **t3** the sonar system receives returns from the emitted message. Between these two times there is an optional wait time **t2**.

FIG. 2B illustrates a multispectral sonar system that acquires multifrequency data **200B**. In particular, three CW signals at widely spaced frequencies 50, 150, 250 kHz are included in a transmitted message. As shown, the signals are arranged to occur during the same time span **t11-t22**. Notably, in some embodiments the signals are of differing durations.

FIG. 2C illustrates a multispectral sonar system that acquires multifrequency data **200C**. In particular, three CW signals at widely spaced frequencies 50, 150, 250 kHz are included in a transmitted message. As shown, the signals are arranged serially such that they occur in time spans **t11-t22**, **t22-t33**, and **t33-t44**. Notably, in some embodiments there are temporal gaps between the signals.

Sonar systems capable of constructing the messages of FIG. 2B-C, ensonifying a target with the message, and processing the returns include multispectral systems that operate at the three frequencies. These sonar systems must segregate the signals at the three frequencies, and bandpass filters or matched filters or their equivalents are typically used for this purpose.

Multispectral sonar systems such as those described above may be distinguished from hyperspectral sonar systems. In particular, hyperspectral sonar systems utilize a broadband signal to excite all frequencies within the band for ensonifying a target and have no need of segregating signals at various frequencies during transmission as was the case for multispectral sonar systems.

Broadband signals are contrasted with narrow band signals. In an embodiment, a signal is a broadband signal when it is not a narrow band signal.

In an embodiment, broadband signals occur in the range of 20 kHz to 1000 kHz and may have bandwidths in the range of 20 kHz to 1000 kHz.

In an embodiment, a signal is a broadband signal when it exceeds 20 kHz and its bandwidth is more than 10% of the center frequency.

In an embodiment, a signal is a broadband signal when a statistically significant difference exists between a) an acoustic return from a target excited by a first frequency in the band and b) an acoustic return from the target excited by a second frequency in the band.

In an embodiment, a signal is a broadband signal when the backscatter strength at first frequency in the band differs by more than a prescribed amount from the backscatter strength at a second frequency in the band. In some embodiments a difference of about 2 dB or more may indicate that the signal is a broadband signal. An example follows.

Consider a message comprising signals S_x in frequency bands B_x and B_y where B_x is the lower of the two frequency bands. Where backscatter signal strengths BS_x and BS_y differ by 2 dB or more, then the signal is a broadband signal.

Any one or more of the above described methods may be used to determine whether a signal is a broadband signal.

FIG. 2D shows a multispectral survey system 200D. The echo sounder system includes a transducer section 220, a transmitter section 250, and a receiver section 270. Some embodiments include an interface section 290 and/or a management section 292.

Notably, N may vary from 2 to the number of different center frequencies to be transmitted. In the embodiment shown, $N=2$, so a message 253 incorporating first and second signals S_{cf1} , S_{cf2} at first and second different center frequencies f_1 , f_2 is used to excite three projectors in a projector array. A receiver having three hardware pipelines and six software pipelines is used to process T hydrophone signals for recovery of echo information specific to each of N frequencies. Note that T is the number of hydrophones and that here, $T=3$.

The transmitter section 250 is for exciting the projector array 230. The section includes a signal generator block 258, a transmit beamformer block 256, a summation block 254, and a power amplifier block 252.

In the signal generator block 258, N signal generators are shown operating at different user selectable center frequencies f_1 , f_2 . In respective beamformers of the beamformer block 256, multiple beams are generated from each signal. In a summation block 254, the beams are combined to produce a summation block output signal 253.

The transducer block 220 includes a projector array 230 and a hydrophone array 240 arranged as a Mills Cross. As shown, there are three projectors 231 in the projector array and three hydrophones 241 in the hydrophone array. In the power amplifier block 252, the summed signal or message 253 is an input to power amplifiers driving respective projectors.

Applicant notes that for convenience of illustration, the projector and hydrophone counts are limited to three. As skilled artisans will appreciate, Mills Cross arrays need not have equal numbers of projectors and hydrophones nor do the quantities of either of these transducers need to be limited to three. For example, a modern multibeam echo sounder might utilize 1 to 96 or more projectors and 64 to 256 or more hydrophones.

The array of T hydrophones 241 is for receiving echoes resulting from the acoustic/pressure waves originating from the projector array 230. The resulting hydrophone signals are processed in the receiver section 270 which includes a hardware pipeline block 272, a software pipeline block 274, a receive beamformer block 276, and a processor block 278.

In the hardware pipelines block 272, each of T hardware pipelines processes a respective hydrophone 241 signal through analog components including an analog-to-digital converter. In the embodiment shown, a hardware pipeline provides sequential signal processing through a first amplifier, an anti-aliasing filter such as a low pass anti-aliasing filter, a second amplifier, and an analog-to-digital converter.

In the software pipelines block 274, each of the T hardware pipeline outputs is processed through N software

pipelines with downconversion and matched filtering. In the embodiment shown, a software pipeline provides sequential signal processing through a mixer (oscillator is not shown for clarity), a bandpass filter, a decimator, and a matched filter.

Communications may occur via communications links between any of the processor block 278, the signal generator block 258, the hardware pipelines block 272, the software pipelines block 274, the and the beamformer block 276. See for example FIG. 2D.

In the receive beamformer block 276, each of N beamformers processes signals. As such, three software pipeline outputs at a first center frequency are processed by a first beamformer and three software pipeline outputs at a second center frequency are processed by a second beamformer. Notably, beamformers may be implemented in hardware or software. For example, multiple beamformers may be implemented in one or more field programmable gate arrays ("FPGA").

In the processor block 278, each of N processors are for processing respective beamformer outputs. Here, a first plurality of beams generated by the first beamformer is processed in a first processor and a second plurality of beams generated by the second beamformer is processed in a second processor. Processor outputs interconnect with a management section 292. Notably, one or more processors may be implemented in a single device such as a single digital signal processor ("DSP") or in multiple devices such as multiple digital signal processors.

Complementary data may be provided via a sensor interface section 290 that is interfaced with a plurality of sensors ES1, ES2, ES3. The sensor interface module may provide sensor data to management section 292 and/or to processors in the processor block 278.

In an embodiment, management section 292 and sensor interface section 290 are provided. The management section includes an interface module 294 and/or a workstation computer 296. The sensor interface section provides for interfacing signals from one or more sensors ES1, ES2, ES3 such as sensors for time (e.g. GPS), motion, attitude, and sound speed.

The management section 292 includes a sonar interface 294 and/or a workstation computer 296. In various embodiments control signals from the management block 292 are used for one or more of making power amplifier block 252 settings (e.g., for array shading), controlling transmit 256 and receive 276 beamformers, selecting software pipeline block 274 operating frequencies, setting set signal generator block 258 operating frequencies, and providing processor block 278 operating instructions.

As shown in FIG. 2D above, the projectors are driven by a signal at discrete frequencies. Returns received by the hydrophones are mixed with respective oscillator signals and the signals, for example the difference signals, are passed through respective band pass and matched filters. Hyperspectral Sonar Systems

FIG. 3A shows a hyperspectral broadband signal 300A. As seen, the bandwidth of the signal is from 100 to 700 kHz, a range of 600 kHz.

Referring again to the transducers of FIG. 1 103, 107, these projectors and hydrophones have a much wider and higher operating band than that used by conventional sonar systems that may also use broadband waveforms. In particular, these projectors and hydrophones may operate in a frequency range of 20 to 1000 kHz. Furthermore, the

spacing of transducer elements in the transmit and receive arrays is selected to allow beams to be formed with 1-degree resolution.

FIG. 3B shows the hyperspectral signal of FIG. 3A used to ensonify a target 300B. In operation, the hyperspectral sonar ensonifies a target with this signal such that the target is excited with all of the frequencies in the band 100 to 700 kHz during time span t11-t22.

FIG. 3C shows the hyperspectral signal of FIG. 3B and exemplary center frequencies of interest 314 (300 kHz, 400 kHz, 600 kHz). Notably, with hyperspectral sonar the center frequencies of interest can be chosen after the data are acquired because data for all of the frequencies in the band are returned.

If the available bandwidth of the sonar equipment is less than that needed for the hyperspectral signal, the sonar may be adapted by aggregating bands. FIGS. 4A-D below illustrate the case of adequate sonar equipment bandwidth while FIGS. 4E-H illustrate the case of aggregated bands to accommodate limited bandwidth.

Hyperspectral Sonar, No Bandwidth Limitation

FIG. 4A shows a hyperspectral broadband signal 400A. As seen, the bandwidth of the signal is from 80 to 260 kHz, a range of 180 kHz.

FIG. 4B shows the hyperspectral signal of FIG. 4A used to ensonify a target 400B. In operation, the hyperspectral sonar signal 404 ensonifies a target such that the target is excited by all of the frequencies in the band 80 to 260 kHz during time span t11-t22.

FIG. 4C shows an embodiment including a single beam hyperspectral sonar system 400C. The sonar system includes a signal generator 458 for constructing a message that is delivered to an amplifier 452 and thereafter to a projector 431. Returns from the message are picked up by a hydrophone 441 and processed. The projector and hydrophone may comprise an integral unit. There may be one or several projectors and/or hydrophones, and one transducer may act as both a projector and a hydrophone.

Processing occurs in a processing pipeline 470. The processing pipeline includes a first pipeline block 472 which may be hardware or software and a second pipeline block 474 which may be hardware or software. In an embodiment, the first block includes hardware and in an embodiment the second block includes software. In an embodiment, the first block consists of hardware and in an embodiment the second block consists of software.

The first pipeline block 472 may include one or more of amplification followed by anti-aliasing which is again followed by amplification and analog to digital conversion.

The second pipeline block 474 may include one or more of mixing followed by bandpass filtering followed by decimation. An oscillator may be connected with the mixer as shown. The first pipeline block 472 feeds the second pipeline block 474 which feeds facilities for processing and storage 410 at interface 412. In various embodiments, the processing and storage facility receives a beam time series via the pipelines.

The processing and storage facilities 410 may include processing, storage, or processing and storage. Processing of the second pipeline 474 output may precede storage. Storage may include the beam time series or data derived from the beam time series. In various embodiments, the storage provides a means to examine the returns at any frequency within the band encompassed by the broadband message emitted by the projector 431.

As shown, the processing and storage facility 410 may include any of a processor 414, a real time output 420, and

a storage section 430. In an embodiment, the processing and storage facility interface 412 receives the output of the second pipeline block 474 and may exchange signals with a sensor interface 490. The sensor interface section provides for interfacing signals from one or more sensors ES1, ES2, ES3 such as sensors for time (e.g. GPS), motion, attitude, and sound speed.

In the processor block 414 calculations may be performed including calculations measuring round trip travel time for the acoustic signal emitted by the projector 431. Processor outputs may interconnect with a management section 435. Notably, one or more processors may be implemented in a single device such as a single digital signal processor ("DSP") or in multiple devices such as multiple digital signal processors.

The management section 435 may include a sonar interface and a workstation computer. In various embodiments control signals from the management section are used for one or more of making power amplifier block 452 settings, selecting software pipeline block 474 operating frequencies, setting signal generator block 458 broadband message particulars, generating signals 458, and providing processing and storage facility 410 instructions. In various embodiments the processor block 414 and management section 435 may share processors.

FIG. 4D shows an embodiment including a multibeam hyperspectral sonar system 400D. While this sonar system is similar to the multispectral sonar system described above, there are important differences. Differences include changes made in the transmitter section 450, the second pipeline block 474, in the beam forming section 476, and in the processor section 478.

In the transmitter section, a single broadband signal is generated 456. This signal is subsequently passed through a beam former 454. The beam former output or a signal derived therefrom is amplified in power amplifier(s) 452. For example, three projectors are shown and each projector is driven by a respective power amplifier.

In the second pipeline block 474, a first difference is the elimination of matched filters. A second difference is the elimination of the multispectral aspect that provided for each hydrophone two mixing/filtering sections to handle returns from a transmitted multispectral message including content at a first frequency and content at a second frequency. In the drawing, a mixer 4741 is followed by a notch filter 4742 and a down converter 4743. An oscillator is for use with each of the mixers 4744. In an embodiment, each pipeline has an oscillator and the oscillator may provide a common oscillator frequency to each mixer. In an embodiment, the oscillator signal is provided by a common oscillator.

In the beam forming section 476, two beam formers were required where the multispectral system transmitted a message with two different frequency components. However, in the hyperspectral system only a single beam former is required to process returns from a broadband message.

In the processing and storage facility 410, a signal interface 412 receives the beam former output and may be interconnected with one or more of a processor 414, a real time unit 420 and a storage unit 430.

Complementary data may be provided via the interface 412. For example, a sensor interface section 490 may provide for interconnection with a plurality of sensors ES1, ES2, ES3. Sensors may include sensors for time, motion, altitude, and sound speed.

In some embodiments, a management section 435 is provided. The management section may include a sonar interface and/or a workstation computer. In various embodi-

ments control signals from the management section are used for one or more of making power amplifier block **452** settings (e.g., for array shading), controlling transmit **454** and receive **476** beamformers, selecting software pipeline block **474** operating frequencies (where broadband signals are subdivided), setting signal generator block **456** signal particulars including a start frequency and bandwidth, and providing processor block **478** operating instructions.

As shown in FIG. 4D above, the projectors are driven by a single broadband source. Returns received by the hydrophones are mixed with respective oscillator signals and the signals are passed through respective band pass filters before reaching the beamformer. No matched filters are used.

Hyperspectral Sonar, Bandwidth Limited

FIG. 4E shows a hyperspectral broadband signal **400E**. As seen, the bandwidth of the signal is from 80 to 260 kHz, a range of 180 kHz. But, as seen this bandwidth is divided into three 60 kHz sub-bands.

FIG. 4F shows a serial hyperspectral signal **400F** incorporated in a message. A low band **4120** of 60 kHz adjoins an intermediate band **4140** of 60 kHz which adjoins a high band **4160** of 60 kHz such that the 180 kHz range is covered. As seen, the sub-bands are transmitted in serial fashion with the low band transmitted in time span **tt1-tt2**, the intermediate band transmitted in time span **tt2-tt3** and the high band transmitted in time span **tt3-tt4**.

FIG. 4G shows a parallel hyperspectral signal **400G** incorporated in a message. A low band **4252** of 60 kHz adjoins an intermediate band **4254** of 60 kHz which adjoins a high band **4256** of 60 kHz such that the 180 kHz range is covered. As seen, the sub-bands are transmitted in parallel fashion in time span **tt11-tt22**.

FIG. 4H shows a modification **400H** to the multibeam hyperspectral sonar system of FIG. 4D that accommodates sub-bands. As seen, the analog to digital converter outputs are passed to branches that are filtered via bandpass filters to accommodate each of the sub-bands. Notably, each branch includes a mixer, a bandpass filter, and a down converter. Oscillators interconnected with the mixers are omitted for clarity.

In this example, there are three sub-bands. Therefore, each analog to digital converter feeds three branches. And, each branch has a bandpass filter that selects a sub-band. Bandpass filters are therefore provided for 80 to 140 kHz, for 140 to 200 kHz, and for 200 to 260 kHz.

After processing in the branches, the output of each analog to digital converter is summed. The summed outputs are then passed to the beamformer.

FIG. 5 shows a mission table **500**. Bottom missions may include bottom classification and/or segmentation, sub-bottom classification and/or segmentation, and improved bottom detection. Other missions may include water column classification and/or segmentation, and object classification. All of these missions may be carried out with broadband messages including either of frequency modulation (FM), noise-like, click, or click train signals.

Bottom classification involves partitioning acoustic returns from the seafloor into discrete groups associated with various sediment or bottom types. Variations in backscatter strength from multiple narrowband frequencies are used to construct a feature set representing sea floor response versus frequency or versus frequency and incident angle. (It should be noted that there is an upper frequency above which backscatter strength ceases to vary with frequency.) These features can then be used independently to segment bottom

types or they can be correlated to expected responses from known sediment types in attempts to characterize the sea floor.

During a bottom classification and/or segmentation mission, a broadband message in the range of 20 to 800 kHz may be used. This message may have a bandwidth of 100 kHz or greater.

Sub-bottom classification is similar to bottom classification in its use of variations in acoustic returns from multiple narrowband frequencies to characterize sea floor types, but its use of lower frequency sounds allows acoustic penetration into the sea floor. Sub-surface classification can characterize sediment types, rocks, or fluids beneath the sea floor.

During a sub-bottom classification and/or segmentation mission, the message frequency will tend toward the lower end of the frequency range. A broadband message in the range of 20 to 100 kHz may be used. This message may have a bandwidth of 40 kHz or less.

During an improved bottom detection mission, a broadband message in the range of 20 to 800 kHz may be used. This message may have a bandwidth of 100 kHz or greater. Dominant or predictable clutter frequencies can be ignored in favor of those with high backscatter strength from a known bottom type.

Water column classification is a method for detecting and identifying aspects of the marine environment between the ocean surface and sea floor. Naturally occurring content in the water column can include animals, plants, dust, and bubbles. Frequency-dependent features of the acoustic backscatter from these objects can be used to identify or infer properties of the features causing the backscatter.

During a water column classification and/or segmentation mission, a broadband message in the range of 20 to 800 kHz may be used. The message may have a bandwidth of 100 kHz or greater.

During a mission to classify objects, for example man-made objects from natural clutter objects, sonar resolution may be insufficient for classification. In such cases, variations in target response from different narrowband frequencies can be provided to automated clustering, segmentation, or classification applications.

During an object classification mission, a broadband message in the range of 20 to 800 kHz may be used. The message may have a bandwidth of 100 kHz or greater. The message may have a bandwidth of 100 kHz or greater.

FIG. 6A shows a hyperspectral sonar method using band pass filters **600A**. In a first step, data acquired by the sonar system including one or more acoustic returns are indexed by ping number, by beam number, and by sample number **602**.

In a second step **603** the data are stored. The data may be used contemporaneously or the data may be used at a later time, for example to evaluate the strength of the returns at various frequencies.

In a third step, a particular ping is selected **604**. In a fourth step, a particular beam is selected **606**. In a fifth step, the time series associated with the particular beam is identified **608**.

In a sixth step, a band pass filter range is selected **610**. In a seventh step, the time series is passed through the band pass filter **612**. In an eight step **614**, the band pass filter output is the frequency content within the pass band.

Notably, the width of the band pass filter may be a fraction of the center frequency. For example, the width of the band pass filter may be 10% of the center frequency. For example, the width of the band pass filter may be 1%, 2%, 3%, 4%,

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5%, 6%, 7%, 8% or 9% of the center frequency. In an embodiment, the width of the band pass filter is 2 kHz or between 2 and 10 kHz.

In a ninth step, a query asks if enough band pass filter ranges have been selected **616**. If so, the process subsequently compares band pass filter outputs in step **620**. If not, the process selects another band pass filter range **618** and subsequently moves to step **612**.

FIG. 6B shows a hyperspectral sonar method using Fourier Transforms **600B**. In a first step, data acquired by the sonar system including one or more of acoustic returns or travel times is indexed by ping number, by beam number, and by sample number **652**.

In a second step **653** the data are stored. The data may be used contemporaneously or the data may be used at a later time, for example to evaluate the strength of the returns at various frequencies.

In a third step a particular ping is selected **654**. In a fourth step, a particular beam is selected **656**. In a fifth step, the time series associated with the particular beam is identified **658**.

In a sixth step, a Fourier transform or fast Fourier transform is used to process the time series **660**. In a seventh step the Fourier transform output is a spectrum of the time series indexed by frequency **662**.

In an eighth step, the spectrum provides the relative strength of the echo at each of multiple quantized frequencies **664**. In a ninth step, the spectrum at the frequency bins of interest are compared **666**.

While various embodiments of the present invention have been described above, it should be understood that they have been presented by way of example only, and not limitation. It will be apparent to those skilled in the art that various changes in the form and details can be made without departing from the spirit and scope of the invention. As such, the breadth and scope of the present invention should not be limited by the above-described exemplary embodiments, but should be defined only in accordance with the following claims and equivalents thereof.

What is claimed is:

1. A hydrographic sonar system comprising:

a multibeam sonar transceiver including a transmitter and a receiver;

the transceiver for use with one projector array and one hydrophone array where the arrays are not arranged in plural rows and plural columns;

the projector array forms one transmit beam that results from a single broadband waveform that excites all frequencies within the frequency band;

the transmit beam being an elongated transmit beam that simultaneously ensonifies an entire swath;

the hydrophone array forms multiple receive beams that intersect the transmit beam, sonar data being acquired at the intersections;

the sonar data, or information derived from the sonar data, processed to recognize signals at selected frequencies; and,

bottom segmentation or characterization indicated by the processed data.

2. A hydrographic sonar system comprising:

a multibeam sonar transceiver including a transmitter and a receiver;

the transceiver for use with one projector array and one hydrophone array where the arrays are not arranged in plural rows and plural columns;

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the projector array forms one transmit beam that results from a single broadband waveform having a bandwidth of 100 kHz or more;

the transmit beam being an elongated transmit beam that simultaneously ensonifies an entire swath;

the broadband waveform excites all frequencies within the band;

the hydrophone array forms multiple receive beams that intersect the transmit beam, sonar data being acquired at the intersections; and,

the sonar data or information derived from the sonar data processed by one or more bandpass filters;

wherein bandpass filter outputs are computed mathematically at two or more bandpass filter frequencies to aid in bottom segmentation or characterization.

3. The sonar system of claim 2 wherein matched filters are not used.

4. The sonar system of claim 2 wherein pulse compression is not used.

5. The sonar system of claim 2 wherein the receiver includes a single beam former.

6. The sonar system of claim 5 wherein plural mixers using respective oscillators are included in the receiver.

7. The sonar system of claim 2 wherein the broadband waveform produces a statistically significant difference between an acoustic return from a target excited by a first frequency in the band and an acoustic return from the target excited by a second frequency in the band.

8. The sonar system of claim 2 wherein the broadband waveform includes a first signal with a first backscatter signal strength and a second signal with a second backscatter signal strength, the difference between the backscatter strengths being 2 dB or more.

9. The sonar system of claim 2 wherein the processed data are divided into multiple subsets and a comparison of the subsets is influenced by band pass filter outputs.

10. A hydrographic sonar survey system comprising:

a plurality of transducers in one projector array and a plurality of transducers in one hydrophone array;

a first projector in the projector array generates an acoustic signal with a frequency or phase that is not independent from a second projector in the projector array;

the projector array configured to form one transmit beam and the hydrophone array configured to form multiple receive beams;

the transmit beam is intersected by the receive beams;

the system configured to include an operating mode in which the plurality of transducers in the projector array emit a broadband waveform, the waveform having a bandwidth of more than 10% of its center frequency;

the hydrophone array configured to receive echoes from scattering centers ensonified by the broadband waveform;

echoes received by the hydrophones result in data derived from the received echoes;

the data are divided into subsets based on frequency; and,

at least two of the filtered subsets are compared, information derived from the comparison for analysis of a waterbody and/or its containment.

11. The sonar system of claim 10 wherein matched filters are not used.

12. The sonar system of claim 10 wherein pulse compression is not used.

13. The sonar system of claim 10 wherein the receiver includes a single beam former.

14. The sonar system of claim 13 wherein plural mixers using respective oscillators are included in the receiver.

15. The sonar system of claim **10** wherein the broadband waveform produces a statistically significant difference between an acoustic return from a target excited by a first frequency in the band and an acoustic return from the target excited by a second frequency in the band.

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16. The sonar system of claim **10** wherein a comparison of the subsets includes analysis of a first signal with a first backscatter strength and analysis of a second signal with a second backscatter strength where the difference between the backscatter strengths is 2 dB or more.

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17. The sonar system of claim **10** wherein a comparison of the subsets is influenced by band pass filter outputs.

* * * * *